Brandon Li

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Education

University of California, Los Angeles (UCLA) Expected June 2028
Bachelor of Science in Computer Science *Los Angeles, CA*

Experience

Software Development Engineer Intern Sep 2026 – Present
Amazon (Amazon Web Services) *Boston, MA*

- Extended Amazon's Remote MCP server with 8 endpoints by integrating an internal work-management service (similar to Jira), enabling hundreds of thousands of employees to execute workflows directly from a chatbot.
- Built specialized AI agents for internal work-management workflows, allowing agent-to-agent (A2A) coordination for custom tasks such as summarization and board updates while handling thousands of production requests.
- Implemented A2A authentication and support for user-mocking agents for secure agent interactions and handoffs.
- Improved production reliability and debuggability by adding logging and over a hundred unit, wiring, and deployment tests, catching possible failures before release and validating end-to-end deployment behavior.

Software Engineering Intern June 2026 – Sep 2026
Google (YouTube) *San Bruno, CA*

- Engineered 10-bit software AV1 HDR video playback for the YouTube Android app via low-level ARM/NEON SIMD vectorization in C++, improving the video quality and playback performance for millions of users.
- Defined per-device video resolution caps (e.g. 1080p) for dav1d software decoding, filtering higher-resolution formats out of the playback selection to prevent thermal throttling and battery drain on lower-end devices.
- Built a Java-based shader effect debug UI and OpenGL video pipeline to accelerate developer iteration speed.
- Integrated custom fragment shaders to allow video effects across thousands of DRM-protected shows and movies.
- Resolved memory leaks, migrated deprecated code, and improved format filtering across ExoPlayer and YouTube.

Software Engineering Intern June 2025 – Sep 2025
VideoLAN *Remote*

- Built a shared audio dynamics core, cutting duplicate code by 20% and binary size by 5% to speed up downloads.
- Added ggml/OpenCV to VLC, ported 3 modules to C++, and upgraded to SAM3 to support text segmentation.
- Designed SAM2 object segmentation and YuNet face detection to be able to track over 100 concurrent targets.
- Implemented 5 different dithering algorithms to allow users to convert videos into GIF files at varying qualities.

Projects

yt-media-storage | *C++, FFmpeg, SIMD, OpenMP, Qt 6, libsodium, Wirehair* Feb 2026 – Present

- Built a C++ tool (920 stars, 110+ forks) that can convert desktop files into YouTube videos for storage.
- Implemented SIMD intrinsics, OpenMP parallelization, and cache optimizations to speed up conversion by 100x.
- Designed a lightweight UI using Qt 6 and added libsodium encryption to enhance usability and security.
- Used CRC and Wirehair error correction to recover data corrupted by lossy video compression algorithms.
- Reached #5 on Hacker News (Y Combinator) in 12 hours, drawing thousands of views and new contributors.
- Published a technical explanation video on YouTube with over 3.3M impressions, educating over 280k viewers.

mcav | *Java, GLSL, libVLC, OpenCV, QEMU, VNC, TeamCity, Gradle* June 2020 – Present

- Built a 15-module, 230K-line Java media framework (130+ stars, 8 forks) whose unified player API streams real-time video from VLC, OpenCV, Chromium, VNC, and QEMU sources across four pluggable render backends.
- Designed a pure-GPU (GLSL fragment shader) video codec with O(1) per-pixel random access that decodes 1080p in 5 ms/frame on integrated GPUs, raising VMAF from 11 to 75 at a lower bitrate than the old dithering method.
- Engineered a low-latency pipeline API with nearly 50 different documented video effects that can be applied.
- Self-hosted a TeamCity CI/CD pipeline that publishes releases to a Maven repository, gated with 4,000+ tests.

Technical Skills

Languages: Java, C/C++, OCaml, JavaScript/TypeScript, Assembly
Frameworks & Libraries: Spring, FFmpeg, OpenCV, libVLC, ExoPlayer, Qt, OpenMP, React, Next.js, Tailwind
Technologies: AWS, AI Agents, MCP, Git, CMake, Gradle, TeamCity, OpenGL, Android, GLSL, AV1, NEON/SIMD
Awards: Dean's Honors List